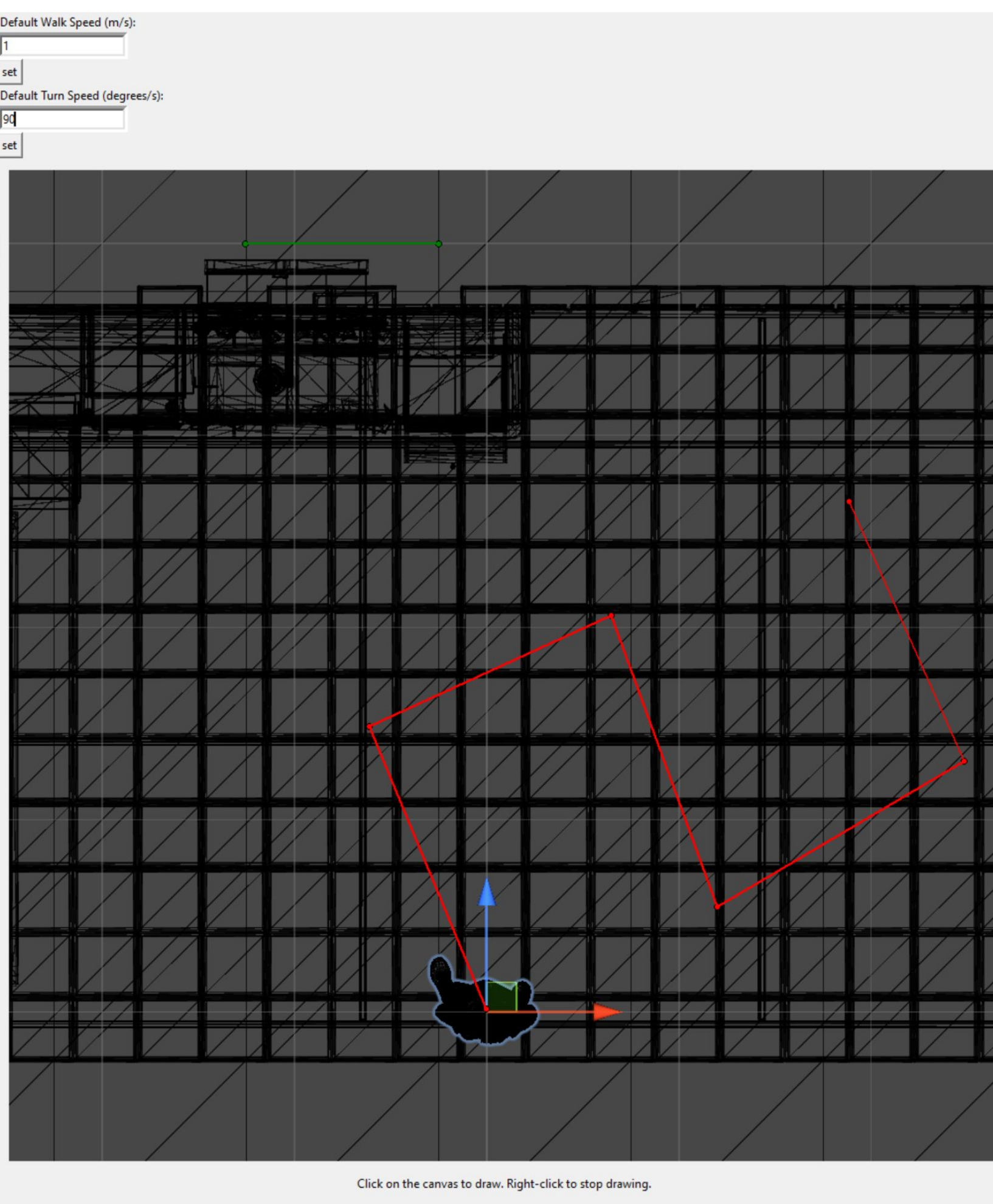


Project Statement

The Amazon Lab126 Clinic project aims to create a software tool to generate simulations of different acoustic environments. This tool would need to accurately digitize human motion and room layouts using 3D meshes, as well as positions per frame. Those meshes would then be fed into Amazon's acoustic simulator.

Trajectory Mapping

Mapping just a simple polyline trajectory requires a surprisingly high number of parameters. A GUI application provides a high level abstraction for human users to quickly specify these parameters. The GUI (left figure) allows clicking points on a canvas to quickly generate the trajectory and printing them to a text file (right figure) for parsing within Unity.



GUI for drawing trajectory
Trajectory labeled in red

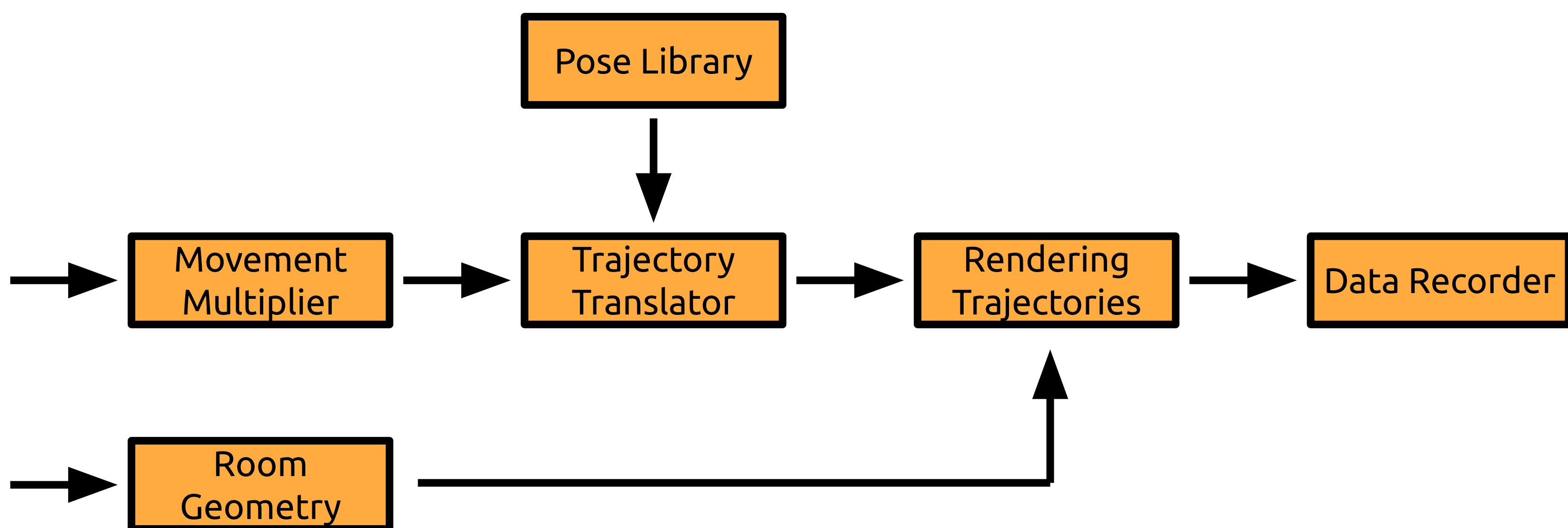
$$T = u(p_0, \theta_0) + \sum_{n=0}^{\infty} f(a_n, v_n, t_n)$$

TurnTrajectory6_ - Notepad
File Edit Format View Help
0,0,2 # idle for 2 seconds.
1,1,1.5 # walk at 1 m/s for 1.5 seconds
1,2,1.5 # walk at 2 m/s for 1.5 seconds
1,1,1.5 # walk at 1 m/s for 1.5 seconds
0,0,1 # idle for 1 second. End. |

High level representation and example of a trajectory.

$u(p, \theta)$ represents the starting position and orientation of the model. $f(a, v, t)$ represents the animation, velocity of movement and duration of movement for one action respectively.
EG: walk at 1 m/s for 1.5s.

Design and Methods



Our design takes two main forms of inputs: the movement and the room geometry. The movement is parsed from a text file. The room geometry is imported as USD file. At the trajectory translator stage, the movement is matched with corresponding poses for the human body model. The entire simulation is then rendered in the game development platform Unity within the specified room geometry. Finally, the data recorder captures a screenshot of the mesh data at every frame. This data can then be sent to Amazon Lab126 for further testing and experimentation with Amazon products. In the first semester, we built the main pathway with the movement through to the data output. In the second semester, we added additional functionality with the room geometry and pose library.

Unity Implementation

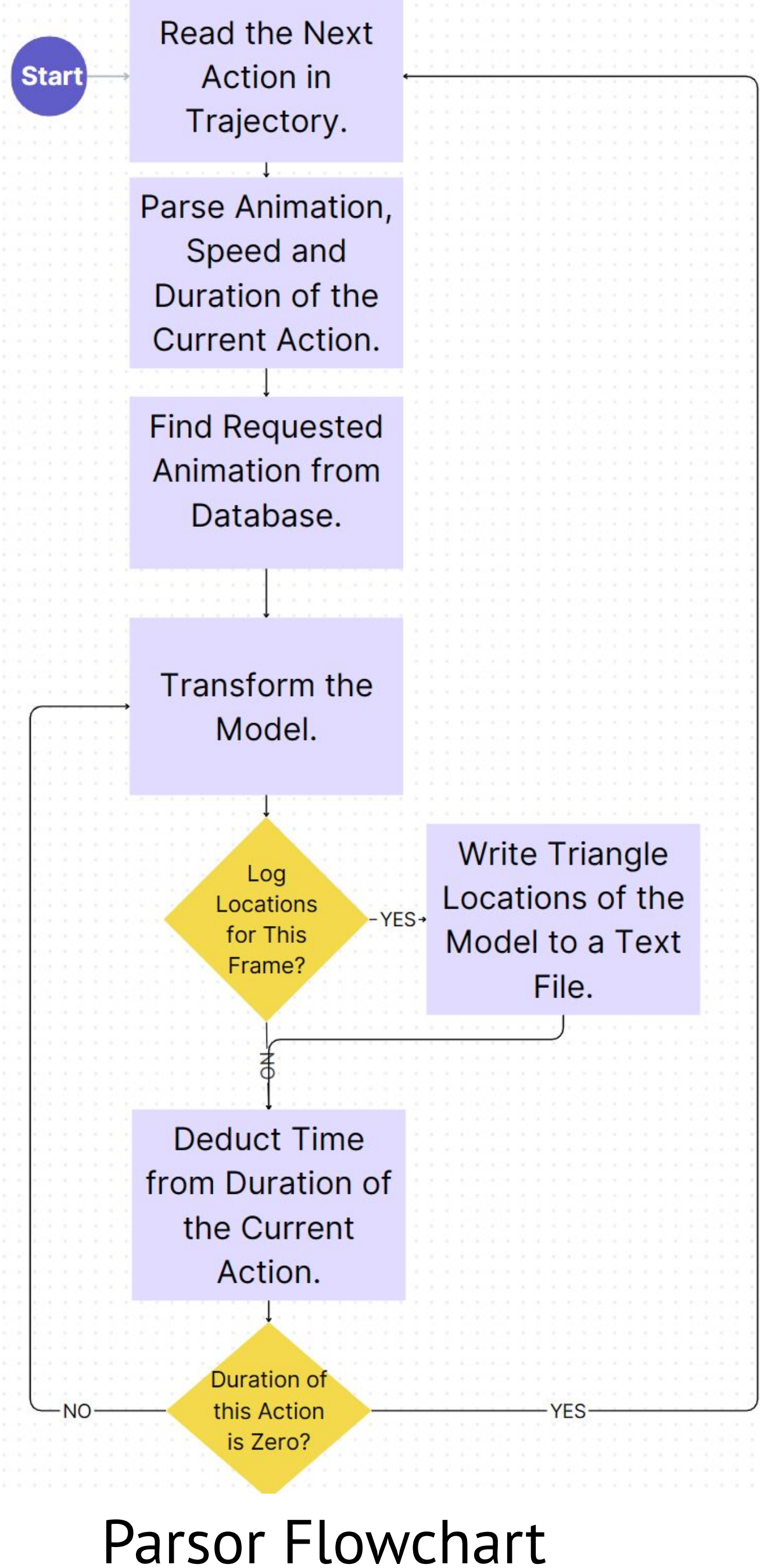


Human mesh in kitchen room geometry

Unity is used to render the simulation inside of a room. The character model is represented as a mesh, which is a combination of vertices and triangles. A level of abstraction called 'bones' characterize the movement of the models.

Rendering Trajectories

Trajectory files are read into the Unity software and parsed according to the flowchart to the right, in real time. A trajectory is a sum of many individual actions. Each action contains three fields: animation, speed and duration. The trajectory, at any time, is a superposition of an animation loop playing at a calibrated speed, on top of a series of translations and rotations of the model determined by speed and duration of the current action.



Parsor Flowchart

Data Output

The data is exported as a combination of vertex identification, x value, y value, and z value. This data include these mesh characterization of the human model, as well the location, position, and movement data. The file can be exported to room geometries as well as other formatting tools. Thus, it can be integrated into future softwares.

```
frame number: 121
format: VertID, Xvalue, Yvalue, Zvalue
0, -12.46304, 2.292101, 0.09146105
1, -12.50087, 2.29528, 0.1388386
2, -12.50461, 2.434901, 0.1619801
3, -12.89432, 2.445089, 0.1907386
4, -12.94178, 2.475636, 0.09824404
5, -12.89172, 2.577835, 0.1897147
6, -12.9382, 2.316774, 0.2362413
7, -12.80645, 2.4035, 0.2507149
8, -12.83174, 2.279427, 0.3228371
9, -12.44992, 2.221571, 0.07209849
10, -12.99253, 2.372467, 0.1062042
11, -12.80237, 2.559525, 0.2398749
12, -12.85782, 2.189354, 0.4372698
13, -12.97258, 2.1968, 0.4107605
14, -12.73678, 2.220887, 0.3317332
```

Example of a single frame export data representation